Project: Worked examples
Title: Thermal expansion of a restrained bar
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

## 1. Scope and method

A bar heated or cooled between two restraints: its free movement, the part closed by a gap before the restraint engages, the restraint force for a rigid or an elastic restraint, the axial stress that results and the movement that actually occurs, with the stress against an allowable.
δ_free = α·L·ΔT. δ_e = sign(δ_free)·max(|δ_free| − g, 0). k_bar = E·A/L. F = δ_e·k_bar (rigid) or δ_e/(1/k_bar + 1/k_s) (elastic restraint). σ = −F/A (compressive on heating). δ = δ_free − F/k_bar.
Ref: Roark's Formulas for Stress and Strain Ch. 15 — Thermal stresses
Ref: Shigley's Mechanical Engineering Design §3-17 — Thermal stresses and strains

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| L | Bar length | 2 | m | drawing: DRG-TX-502 rev A | 0.01 … 1000 m |
| A | Cross-section area | 500 | mm^2 | drawing: DRG-TX-502 rev A | 1 … 1000000 mm^2 |
| E | Modulus of elasticity | 200 | GPa | drawing: DRG-TX-502 rev A | 1 … 500 GPa |
| alpha | Coefficient of thermal expansion | 0.000012 | 1/K | drawing: DRG-TX-502 rev A | 1e-7 … 0.0001 1/K |
| dT | Temperature change (positive heating) | 60 | K | spec: LOAD-SCHEDULE-01 | -300 … 1000 K |
| g | Gap closed before the restraint engages | 0 | mm | drawing: DRG-TX-502 rev A | 0 … 1000 mm |
| k_s | Restraint stiffness (leave empty for a rigid restraint) | 100 | kN/mm | spec: SPEC-MECH-03 design criteria | 0.001 … 10000000 kN/mm |
| sigma_allow | Allowable stress | 150 | MPa | spec: SPEC-MECH-03 design criteria | 1 … 2000 MPa |

## 3. Assumptions

A1. The expansion coefficient is constant over the temperature change and the bar stays elastic. (Linear thermoelasticity.)
A2. The bar is uniformly heated and does not buckle under the compressive force. (Check buckling separately (struct.column-buckling).)

## 4. Working

```
[9] delta_free = alpha * L * dT                    Shigley §3-17
= 0.000012 1/K × 2 m × 60 K
= 1.44 mm
```

Free thermal movement (positive on heating)

[10] delta_e = delta_free - g
= 1.44 mm − 0 mm
= 1.44 mm
Movement after the gap closes (engaged movement)
[11] k_bar = E * A / L
= 200 GPa × 500 mm² / 2 m
= 50 kN/mm
Axial stiffness of the bar
[12] F = delta_e / (1 / k_bar + 1 / k_s) Roark Ch. 15
= 1.44 mm / (1 / 50 kN/mm + 1 / 100 kN/mm)
= 48000 N
Restraint force (elastic restraint in series with the bar)
[13] sigma = -F / A
= −48000 N / 500 mm²
= -96 MPa
Axial stress (negative compressive)
[14] delta = delta_free - F / k_bar
= 1.44 mm − 48000 N / 50 kN/mm
= 0.48 mm
Actual movement
[15] u = abs(sigma) / sigma_allow
= abs(-96 MPa) / 150 MPa
= 0.64
Stress utilisation

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| delta_free | 1.44 mm | Free movement |
| delta_e | 1.44 mm | Engaged movement |
| k_bar | 50 kN/mm | Bar stiffness |
| F | 48000 N | Restraint force |
| sigma | -96 MPa | Axial stress |
| delta | 0.48 mm | Actual movement |
| u | 0.64 | Stress utilisation |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| PASS | Thermal stress against allowable | u = 0.64 | u ≤ 1 | 0.64 | |

## 7. Applicability

No warnings.

## 8. Verification

Template mech.thermal-expansion@1.0.0 Content hash 63e435d7263ca8d1 Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0